

Unit 5

Unity ML Agents components

- Learning env. - contains the scene & game character
- Python Low Level API - low-level Python interface for interacting w/ & manipulating the env.
- External Communication - connects the learning env. w/ the low level python API
- Python trainers - RL algorithms made w/ Python
- Gym wrapper - encapsulates the RL env.
- PettingZoo wrapper - multi-agents version of gym
- The agent component - the actor, training done by maximizing the policy. Policy = brain
- The academy - orchestrates agents & the decision making processes, handles API requests
 - ↳ the academy makes sure that the agents are in sync. i.e. collecting observations & taking actions at the same time.
- The reward engineering problem - having too complex of a reward function will cause the agent to behave in a certain way → a simpler one may allow for the discovery of better strategies.

- the sparse rew. problem - most rewards are set to 0, therefore contain no information, w/o getting a rew. for a while, the agent cannot learn
- the extrinsic rew. func. is handmade - not Scalable to large envs.
- curiosity is an intrinsic rew. func. that rewards the agent for exploring new traj.

$$IR = || \text{predicted}(S_{t+1}) - S_{t+1} ||$$

prediction error of S_{t+1} given S_t and a_t

↓ IR = familiar state, easy to predict next one

↑ IR = unfamiliar state, hard to predict next.

Curiosity will be higher in unexplored env. so it will reward the agent for exploring.